

Experiment 'p_6_2_b_shift_right_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_6_2_b_shift_right_seq

Natural language statement: null

Method signature: p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `result[0] == inputs[|inputs| - 1]`
- `forall i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]`

Requires

- `|inputs| >= 1`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

requires `|inputs| >= 1`, ensures `|result| == |inputs|`, ensures `result[0] == inputs[|inputs| - 1]`, ensures forall `i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNhipB7BptYkLBzGy0UabHZcnWNl

Model: gpt-5.5-2026-04-23

Created at: 1780661562

Finish reason: stop

Usage: 348 tokens in, and 570 tokens out

```
{
  var n := |inputs|;
  result := seq(n, i => if i == 0 then inputs[n - 1] else inputs[i - 1]);
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_6_2_b_shift_right_seq/program.dfy(8,64): Error: index out of range
|
8 | result := seq(n, i => if i == 0 then inputs[n - 1] else inputs[i - 1]);
| ^
```

Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the Dafny body, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNiDW69Q9uQwmhdIbHtKB5sm72w2

Model: gpt-5.5-2026-04-23

Created at: 1780661593

Finish reason: stop

Usage: 543 tokens in, and 342 tokens out

```
{
  var n := |inputs|;
  result := [inputs[n - 1]] + inputs[..n - 1];
}
```

Final Program

```
method p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)
  requires |inputs| >= 1
  ensures |result| == |inputs|
  ensures result[0] == inputs[|inputs| - 1]
  ensures forall i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]
{
  var n := |inputs|;
  result := [inputs[n - 1]] + inputs[..n - 1];
}
```

Total Token Usage

Input tokens: 891

Output tokens: 912

Reasoning tokens: 805

Sum of ‘total tokens’: 1803

Experiment Timings

Overall Experiment started at 1780661561695, ended at 1780661622356, lasting 60661ms (60.66 seconds)

Iteration #1 started at 1780661561696, ended at 1780661593561, lasting 31865ms (31.87 seconds)

Iteration #2 started at 1780661593562, ended at 1780661622356, lasting 28794ms (28.79 seconds)